
Nonparametric Dyadic Quantile Regression

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Why dyads, and why quantiles, in an era of extremes?

Dyadic data are everywhere in macro-finance

- ▶ Bilateral trade, cross-border portfolio holdings, financial flows, risk sharing, policy diffusion
 - ▶ Outcome Y_{ij} lives on the *edge* from i to j ; covariates live on the *nodes*
 - ▶ Unobserved node effects (e.g. multilateral resistance) hit **every** dyad involving that node
- ⇒ ignoring it understates uncertainty

Means are not enough

- ▶ Bilateral positions are extremely heavy-tailed: median \$0.56bn, mean \$22.9bn, max \$1.98tn
- ▶ Interest often lies in the **tails**: the largest exposures drive contagion and stability concerns
- ▶ Conditional mean is sensitive to outliers and hides distributional heterogeneity

This paper

Local linear **nonparametric quantile regression** for **directed dyadic data**, with a rate-adaptive limit theory, a feasible variance estimator and a dyad-aware bandwidth selector.

Where this sits in the literature

- ▶ **Parametric dyadic regression:** OLS with dyadic-robust SEs (Cameron et al., 2011; Tabord-Meehan, 2019); moments (Bickel et al., 2011); likelihood / PPML (Santos Silva & Tenreyro, 2006; Graham, 2020a, 2024)
 - Misspecification $\Rightarrow$ biased estimates and invalid inference
- ▶ **Nonparametric dyadic methods:** kernel density (Graham, Niu & Powell, 2024; Chiang & Tan, 2023; Cattaneo, Feng & Underwood, 2024); conditional mean (Graham, Niu & Powell, 2021)
- ▶ **Nonparametric quantile regression:** i.i.d. and weakly dependent data (Yu & Jones, 1998; Cai & Xu, 2008; Qu & Yoon, 2015; Li, Li & Li, 2021)

Gap

To our knowledge, no nonparametric quantile regression theory for *directed* dyadic data.

Contributions

- 1. Dimension reduction from dependence.** Under non-degeneracy the relevant dimension is $d_X = \dim(X_i)$, not $d_W = 2d_X$: the Hájek projection has a “partial-mean” structure (Newey, 1994).
- 2. Rate-adaptive CLT.** One standardised normal limit covering both the non-degenerate rate $(nh^{d_X})^{-1/2}$ and the degenerate rate $(n^2h^{d_W})^{-1/2}$.
- 3. Feasible inference.** Easy-to-implement plug-in variance estimator using sample splitting by nodes.
- 4. Practice.** Leave-two-node-out (LTNO) cross-validation; Monte Carlo; application to IMF CPIS bilateral portfolio holdings.

Setup: the conditionally independent dyad (CID) model

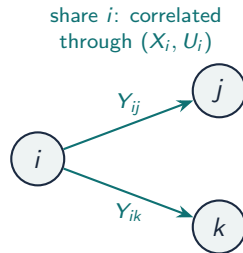
Nodes $i = 1, \dots, n$ drawn from a superpopulation;
directed outcomes

$$Y_{ij} = h(X_i, X_j, U_i, U_j, V_{ij}), \quad 1 \leq i \neq j \leq n$$

- ▶ X_i observed, U_i latent node features, both i.i.d. over i
- ▶ V_{ij} i.i.d. dyad shocks, independent of $\{X_i\}, \{U_i\}$
- ▶ h (the graphon) unknown and possibly nonlinear
- ▶ $W_{ij} = (X_i^\top, X_j^\top)^\top$, so $d_W = 2d_X$

Implications: relative exchangeability, and

- ▶ disjoint indices $\Rightarrow$ independent
- ▶ shared index $\Rightarrow$ possibly correlated



Target and estimator

Conditional τ -quantile of the dyadic outcome:

$$Q_\tau(w) = \inf\{y : F_{Y|W}(y | w) \geq \tau\} = \arg \min_y \mathbb{E}[\rho_\tau(Y_{ij} - y) | W_{ij} = w]$$

with check loss $\rho_\tau(z) = z(\tau - \mathbb{I}\{z < 0\})$.

Local linear dyadic quantile regression (Fan & Gijbels, 1996):

$$(\hat{a}, \hat{b}) = \arg \min_{a,b} \frac{1}{n(n-1)} \sum_{1 \leq i \neq j \leq n} \rho_\tau(Y_{ij} - a - (W_{ij} - w)^\top b) K_h(W_{ij} - w)$$

$$\hat{Q}_\tau(w) = \hat{a}, \quad \hat{Q}'_\tau(w) = \hat{b}, \quad K_h(\cdot) = h^{-d_w} K(\cdot/h)$$

- Looks like the i.i.d. loss (Yu & Jones, 1998) — the difference is entirely in the dependence across the $n(n-1)$ summands
- Local linear: bias reduction, minimax efficiency, boundary adaptivity

Assumptions (brief)

- A1** $Q_\tau(\cdot)$ unique, bounded, twice continuously differentiable
- A2** f_X continuous and positive at $x_o, x_\bullet$; $f_{Y|W}$ bounded, jointly continuous near $(Q_\tau(w), w)$, and $f_{Y|W}(Q_\tau(w) | w) > 0$
- A3** (i) compactly supported symmetric kernel; (ii) $h \rightarrow 0, nh^{d_X} \rightarrow \infty$; (iii) $h^2 = O(\sigma_n(w))$
- ▶ A3(ii) suffices for **consistency**; A3(iii) is needed only for the **CLT**
 - ▶ A3(iii) holds at the MSE-optimal rate in either regime, and under undersmoothing
 - ▶ Example: multiplicative gravity model $Y_{ij} = e^{g(X_i, X_j)} \zeta_{ij}$, $\zeta_{ij} = e^{U_i + U_j} \eta_{ij}$ gives $Q_\tau(w) = e^{g(w)} q_\tau(\zeta)$ — smooth and *non-degenerate*

Main result: rate-adaptive asymptotic normality

Define $\psi_\tau(z) = \mathbb{I}\{z \leq 0\} - \tau$, $R_{ik} = Y_{ik} - Q_\tau(W_{ik})$, and

$$T_{ik}^\circ(w) = \frac{1}{2} [\psi_\tau(R_{ik})K_h(W_{ik} - w) + \psi_\tau(R_{ki})K_h(W_{ki} - w)]$$

$$\sigma_n^2(w) = \underbrace{\frac{\nu_\tau(w)}{nh^{d_X}}}_{\text{node level}} + \underbrace{\frac{\nu_\tau^\dagger(w)}{n(n-1)h^{d_W}}}_{\text{dyad level}}, \quad \nu_\tau(w) = \frac{4h^{d_X} \text{Cov}(T_{ik}^\circ, T_{jk}^\circ)}{(f_{Y|W} f_W(w))^2}, \quad \nu_\tau^\dagger(w) = \frac{\tau(1-\tau) \int K^2}{f_{Y|W}^2 f_W(w)}$$

($f_{Y|W}$ evaluated at $(Q_\tau(w) | w)$)

Theorem 1

Under A1–A3 and either (a) $\nu_\tau(w)$ bounded away from zero, or (b) $\{R_{ij}\}$ i.i.d. and independent of $(\{X_i\}, \{U_i\})$ so $\nu_\tau(w) = 0$:

$$\sigma_n^{-1}(w) \left[\widehat{Q}_\tau(w) - Q_\tau(w) - b_\tau(w)h^2 \right] \xrightarrow{d} N(0, 1), \quad b_\tau(w) = \frac{1}{2} \int s^\top \frac{\partial^2 Q_\tau(w)}{\partial w \partial w^\top} s K(s) ds$$

If $h^2 = o(\sigma_n(w))$ the bias term may be dropped. Under A1, A2, A3(i)(ii), $\widehat{Q}_\tau(w)$ is consistent.

Two regimes, one theorem

	Non-degenerate $\nu_\tau(w) \neq 0$	Degenerate $\nu_\tau(w) = 0$
Source of variance	shared node effects	dyad-level shocks only
Std. deviation	$(nh^{d_X})^{-1/2}$	$(n(n-1)h^{d_W})^{-1/2}$
Effective sample size	n	$n(n-1)$
Effective dimension	d_X	$d_W = 2d_X$
Optimal bandwidth	$h \asymp n^{-1/(d_X+4)}$	$h \asymp n^{-1/(d_X+2)}$
Pointwise MSE	$n^{-4/(d_X+4)}$	$n^{-4/(d_X+2)}$
Example	$Y_{ij} = m(X_i, X_j) + U_i + U_j + V_{ij}$	$Y_{ij} = m(X_i, X_j) + V_{ij}$

- Non-degenerate optimal bandwidth: $h_{\text{opt}} = (d_X \nu_\tau(w) / 4b_\tau^2(w))^{1/(d_X+4)} n^{-1/(d_X+4)}$
- The standardisation by $\sigma_n(w)$ is **rate adaptive**: no need to know the regime in advance

Intuition: why n and d_X , not n^2 and d_W ?

Score as a U-statistic, Hoeffding-type decomposition:

$$\begin{aligned}\Pi_{1,1} &= \frac{2}{n(n-1)} \sum_{i < j} T_{ij}(w) \\ &= \underbrace{\frac{2}{n} \sum_{i=1}^n \{\mathbb{E}[T_{ij} | X_i, U_i] - \mathbb{E} T_{ij}\}}_{\text{Hájek projection: } O_P((nh^{d_X})^{-1/2})} + \underbrace{\Pi_{1,1,2}}_{\text{dyad level}} + \underbrace{\Pi_{1,1,3}}_{\text{degenerate, negligible}}\end{aligned}$$

- ▶ The projection conditions on node i only and *integrates out the partner's covariate* $\Rightarrow$ “partial mean” smoothing over d_X coordinates
- ▶ Covariances from the $O(n^3)$ index-sharing pairs are each $O(h^{-d_X}) \Rightarrow$ variance $O((nh^{d_X})^{-1})$
- ▶ When node effects vanish, the projection is zero and the dyad-level term takes over
- ▶ CLT: martingale-difference array of length $n + n(n-1)/2$; Lyapunov conditions checked in both regimes (following Graham, Niu & Powell, 2024)

Why A3(iii) matters: a cautionary example

- ▶ Q_τ is only C^2 : higher-order bias is $o(h^2)$ and the linearisation remainder is $o_P(h^2 + \sigma_n(w))$
- ▶ A3(iii), $h^2 = O(\sigma_n(w))$, makes both $o_P(\sigma_n(w))$

Counterexample without A3(iii)

$d_X = 1$, degenerate model, $\tau = 1/2$, $V_{ij} \sim \text{Unif}(-1, 1)$, m quartic near $w = (0, 0)^\top$, flat covariate density.

- ▶ $b_\tau(w) = 0$, smoothing bias of exact order h^4 , while $\sigma_n(w) \asymp (nh)^{-1}$
- ▶ $h = n^{-1/10}$ satisfies A3(ii), yet bias/s.e. $\rightarrow \infty$ at rate $n^{1/2}$

A3(iii) is mild: it holds at both MSE-optimal rates (where $h^2 \asymp \sigma_n(w)$) and under undersmoothing.

Estimating the dyad-level component $\nu_{\tau}^{\dagger}(w)$

Need $f_W(w)$ (standard dyadic KDE) and $f_{Y|W}(Q_{\tau}(w) | w)$. Use $\mathbb{E}[k_{b_1}(Y_{ij} - Q_{\tau}(w)) | W_{ij} = w] \approx f_{Y|W}(Q_{\tau}(w) | w)$ (Fan, Yao & Tong, 1996).

Sample splitting by nodes (Li, Li & Li, 2021)

- ▶ M_1 : dyads among nodes $1, \dots, n_1$; M_2 : dyads among nodes $n_1 + 1, \dots, n$; $n_1 = \lfloor n/2 \rfloor$
- ▶ Estimate $\hat{Q}_{\tau}^{M_1}(w)$ on M_1 — independent of M_2 (disjoint node sets)

$$\hat{f}_{Y|W}(Q_{\tau}(w)) = \frac{\sum_{n_1 < i \neq j \leq n} k_{b_1}(Y_{ij} - \hat{Q}_{\tau}^{M_1}(w)) H_{b_2}(W_{ij} - w)}{\sum_{n_1 < i \neq j \leq n} H_{b_2}(W_{ij} - w)}$$

- ▶ **Proposition 1:** consistent if $b_1, b_2 \rightarrow 0$, $nb_1 \rightarrow \infty$, $nb_2^{d_W} \rightarrow \infty$, $h^2 + (nh^{d_X})^{-1/2} = o(b_1)$
- ▶ Plug in: $\hat{\nu}_{\tau}^{\dagger}(w) = \tau(1 - \tau) \int K^2 / [\hat{f}_{Y|W}^2 \hat{f}_W(w)]$

Estimating the node component $\nu_\tau(w)$

Key identity from the proof:

$$\text{Cov}(T_{ik}^\circ(w), T_{jk}^\circ(w)) = \text{Var}(A_k^\circ(w)), \quad A_k^\circ(w) = \mathbb{E}[T_{k1}^\circ(w) | X_k, U_k]$$

- Infeasible: $\frac{1}{n} \sum_k [A_k^\circ(w)]^2$ is ratio-consistent
- Feasible: replace R_{ij} by $\hat{R}_{ij} = Y_{ij} - \hat{Q}_\tau^{M_1}(w)$, average over partners in M_2 :

$$\hat{A}_k^\circ(w) = \frac{1}{n - n_1 - 1} \sum_{n_1 < j \leq n, j \neq k} \hat{T}_{jk}^\circ(w), \quad \widehat{\text{Cov}} = \frac{1}{n - n_1} \sum_{k=n_1+1}^n [\hat{A}_k^\circ(w)]^2$$

$$\hat{\nu}_\tau(w) = \frac{4h^{d_x} \widehat{\text{Cov}}}{[\hat{f}_{Y|W}(Q_\tau(w)) \hat{f}_W(w)]^2}$$

- Uses only node-level averages of quantile scores — cheap to compute

In practice: which variance component?

An estimated node component is positive even when its population value is zero.

Diagnostic (Appendix C):

1. Noise-adjusted covariance: subtract $\hat{s}_k^2(w)/(n - n_1 - 1)$ from each $[\hat{A}_k^\circ(w)]^2$, truncate at zero
2. Evaluate $\hat{\nu}_{\tau,\text{adj}}(w; h)$ at $h = \gamma \hat{h}_{\text{cv}}$, $\gamma \in \{0.75, 1, \dots, 2\}$
3. Regress $\log \hat{\nu}_{\tau,\text{adj}}(w; h)$ on $\log h$

Benchmark slopes

Non-degenerate: ≈ 0 (signal $\propto h^{-d_X}$)

Degenerate: ≈ 2 (heuristic off-centre term)

Cutoff (here $d_X = 1$): slope $< 1 \Rightarrow$ node component

$$\hat{\sigma}_n^2(w) = l_\tau \frac{\hat{\nu}_\tau(w)}{n \hat{h}_{US}^{d_X}} + (1 - l_\tau) \frac{\hat{\nu}_\tau^\dagger(w)}{n(n-1) \hat{h}_{US}^{d_W}}$$

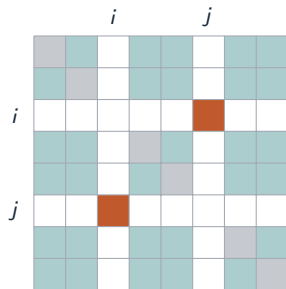
A practical diagnostic, not a formal test; theory for the switching rule is open.

Bandwidth: leave-two-node-out (LTNO) cross-validation

- For each unordered pair $\{i, j\}$: validate on (i, j) and (j, i) ; train on dyads touching *neither* i nor j

$$CV_{\tau}(h) = \frac{1}{n(n-1)} \sum_{\{i,j\}} \sum_{(k,l) \in \{(i,j), (j,i)\}} \rho_{\tau}(Y_{kl} - \hat{Q}_{\tau, -\{i,j\}, h}(W_{kl}))$$

- **Why not leave-one-dyad-out?** Training dyads (i, k) , (k, j) still reveal U_i, U_j ; small h exploits this $\Rightarrow \hat{h}$ biased downward under non-degeneracy
- LTNO leaves no footprint of $U_i, U_j \Rightarrow$ honest out-of-sample loss in both regimes
- Formal optimality: future work



sender $\times$ receiver

■ training ■ validation □ dropped

Monte Carlo design

- ▶ $X_i \sim \text{Unif}(0, 1)$; asymmetric (directed) regression function

$$m(x_o, x_\bullet) = \log(1 + 2x_o + x_\bullet) + 0.5 x_o x_\bullet$$

- ▶ **Non-degenerate:** $Y_{ij} = m(X_i, X_j) + U_i + U_j + V_{ij}$, $U_i \sim N(0, \frac{1}{4})$, $V_{ij} \sim N(0, \frac{1}{2})$
- ▶ **Degenerate:** $Y_{ij} = m(X_i, X_j) + V_{ij}$, $V_{ij} \sim N(0, 1)$
- ▶ Same $Q_\tau(w) = m(w) + \Phi^{-1}(\tau)$ in both, unit error variance
- ▶ Heavy tails: standardised $t(5)$ errors via $\sqrt{3/5} F_5^{-1}(\Phi(G_{ij}))$, preserving each dependence structure
- ▶ $n \in \{15, 30, 60\}$ (210, 870, 3,540 dyads); $\tau \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$; 1,000 replications
- ▶ Product Gaussian kernel; LTNO bandwidth

Theory predicts ($d_X = 1$): doubling n divides IMSE by $2^{4/5} \approx 1.74$ (non-degenerate) vs. $2^{4/3} \approx 2.52$ (degenerate)

Estimation accuracy: IMSE ($\times 1000$)

τ	Gaussian			Standardised $t(5)$		
	$n = 15$	$n = 30$	$n = 60$	$n = 15$	$n = 30$	$n = 60$
<i>Non-degenerate</i>						
0.10	177.76	88.57	45.38	193.93	98.34	52.66
0.50	139.03	73.30	39.35	99.18	48.14	26.26
0.90	189.23	90.73	43.51	192.30	98.86	51.61
<i>Degenerate</i>						
0.10	62.11	18.22	5.27	72.08	21.34	5.86
0.50	33.75	9.95	3.09	23.38	7.04	2.09
0.90	63.28	18.27	5.44	72.35	21.20	6.15

Selected quantiles; full table in the paper (Table 1).

- Median, $n=30 \rightarrow 60$: contraction ≈ 1.9 (non-deg.) vs. ≈ 3.2 (deg.) — two distinct rates, as predicted
- Non-degenerate/degenerate IMSE ratio grows from $\approx 3\text{--}4$ ($n=15$) to $\approx 8\text{--}13$ ($n=60$)
- Heavy tails: lower IMSE at central quantiles, **higher in the tails** ($\tau = 0.1, 0.9$)

Inference: coverage of nominal 95% intervals

τ	$n = 15$	$n = 30$	$n = 60$
<i>Non-degenerate</i>			
0.10	85.9	87.4	88.9
0.25	90.0	91.2	91.2
0.50	91.9	92.1	92.1
0.75	87.6	88.0	91.3
0.90	83.4	87.5	87.7
<i>Degenerate</i>			
0.10	93.4	96.1	96.7
0.25	97.1	97.9	96.4
0.50	97.1	97.3	97.1
0.75	96.4	98.1	97.9
0.90	92.4	95.5	96.1

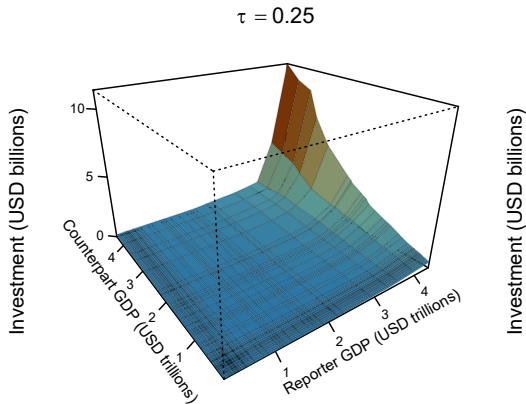
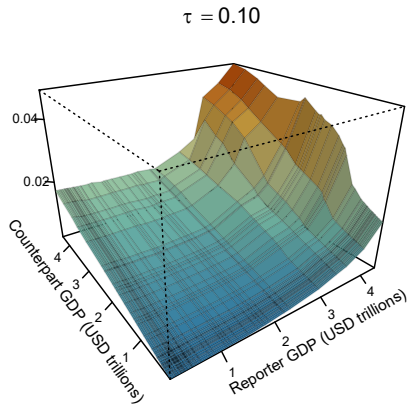
- ▶ $w = (0.5, 0.5)^\top$, Gaussian designs
- ▶ Undersmoothing $\hat{h}_{US} = \hat{h}_{CV} n^{-1/20}$ so that $h^2 = o(\sigma_n(w))$
- ▶ Density bandwidths by LTNO likelihood CV
- ▶ **Degenerate:** close to nominal; average 95.3%, 97.0%, 96.8%
- ▶ **Non-degenerate:** under-coverage but improving; average 87.8% $\rightarrow$ 89.2% $\rightarrow$ 90.2%, non-decreasing in n at every τ
- ▶ Tails are hardest — larger networks help most there

Application: cross-border portfolio holdings

- ▶ **IMF Coordinated Portfolio Investment Survey (CPIS), 2023:** year-end holdings of equity and debt securities by residents of reporter i in issuers of counterpart j
- ▶ **Covariate:** nominal 2023 GDP (World Bank WDI), merged on reporter and counterpart sides — no bilateral controls
- ▶ **Sample:** $n = 51$ economies with $\text{GDP} \geq \$100\text{bn} \Rightarrow 2,550$ candidate dyads; 388 unavailable, 124 zero $\Rightarrow$ **2,038 positive positions**
- ▶ Natural scales, **no log transformation:** Y_{ij} in USD bn, X_i in USD tn

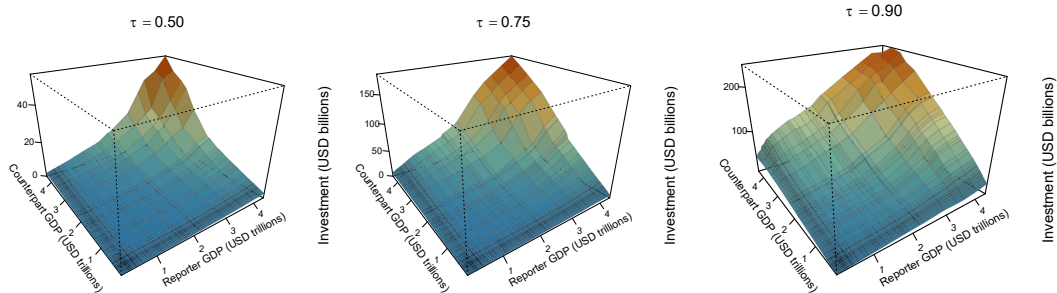
	n	Min	P25	Median	Mean	P75	Max
Y_{ij} (USD bn)	2038	0.000	0.039	0.556	22.858	5.671	1984.535
X_i (USD tn)	51	0.102	0.346	0.516	1.890	1.677	27.292

Estimated conditional quantile surfaces (I)



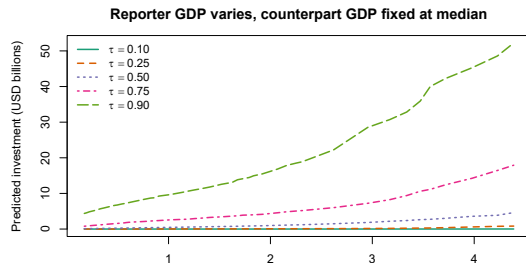
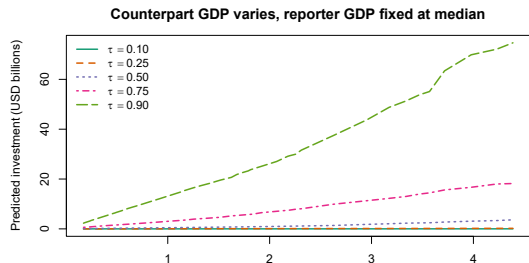
- ▶ $\hat{Q}_\tau(x_o, x_\bullet)$ in USD bn; reporter GDP x_o , counterpart GDP $x_\bullet$; LTNO bandwidth for each τ
- ▶ Flat when both economies are small, rising convexly when both are large

Estimated conditional quantile surfaces (II)



- Scale moves from sub-billion ($\tau = 0.10$) to hundreds of billions ($\tau = 0.90$); **convexity persists at every quantile**

Median-fixed slices: the upper tail fans out



- ▶ Other side's GDP fixed at the median; varying side from 5th to 95th GDP percentile
- ▶ Curves ordered in τ at every point (monotone estimated quantile process)
- ▶ Counterpart slice, $\tau = 0.90$: from roughly \$2bn to roughly \$75bn, convex in partner size; lower quantiles essentially flat on this scale

Quantile-specific reporter-counterpart asymmetry

Reporter	Counterpart	$\tau = 0.10$	0.25	0.50	0.75	0.90
Low	Medium	0.0017	0.0169	0.174	1.194	5.716
Medium	Medium	0.0017	0.0208	0.253	1.621	6.875
High	Medium	0.0024	0.0631	0.884	4.027	15.009
Medium	Low	0.0010	0.0118	0.183	1.120	4.504
Medium	High	0.0068	0.0895	0.868	6.068	24.158
High	High	0.0071	0.1682	2.190	13.860	40.729

USD bn. Low/Medium/High = 20th/50th/80th GDP percentiles (\$0.34, \$0.52, \$1.85tn). Subset of Table 5.

Low → High multiplier

Reporter: 1.4, 3.7, **5.1**, 3.4, 2.6

Counterpart: **6.5**, **7.6**, 4.8, 5.4, 5.4

- ▶ Reporter effect peaks at the **median**
- ▶ Counterpart effect peaks in the **lower tail**
- ▶ A single conditional-mean equation cannot reveal this

Conclusions

- ▶ Local linear quantile regression for directed dyadic data within the CID framework
- ▶ **Rate-adaptive CLT**: convergence depends on whether the network-generating process is degenerate
 - non-degenerate: effective sample n , dimension d_X ; degenerate: $n(n-1)$ and d_W
- ▶ Feasible plug-in variance via node-based sample splitting; LTNO cross-validation for bandwidths
- ▶ Simulations confirm the two contraction rates; good coverage under degeneracy, improving with n otherwise
- ▶ CPIS application: convex, heavy-tailed quantile surfaces without logs, and a quantile-specific reporter-counterpart asymmetry

Open questions

Theory for LTNO optimality • formal justification of the variance-component switching rule • uniform inference across w and τ • richer (bilateral) covariates and sparse networks

Thank you

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Backup: proof architecture

1. **Reparametrise** $\phi = (a - Q_\tau(w), h(b - Q'_\tau(w)))^\top$; Knight's (1998) identity gives $L_\tau(\phi) = \phi^\top \Pi_1 + \Pi_2$
2. **Quadratic part** (Lemma 3): $\Pi_2 = \frac{1}{2} f_{Y|W}(Q_\tau(w) | w) f_W(w) \phi^\top \Lambda(K) \phi + o_P(1)$
3. **Linear part**: $\Pi_1 = \Pi_{1,1}$ (score, mean zero) + $\Pi_{1,2}$ (bias, Lemma 2: $\propto h^2 b_\tau(w)$)
4. **Consistency**: convexity lemma (Pollard, 1991)
5. **Bahadur representation** (Lemma 5, Hjort & Pollard, 1993):
 $\widehat{Q}_\tau(w) - Q_\tau(w) = -e_1^\top V^{-1} \Pi_1 + o_P(h^2 + \sigma_n(w))$
6. **CLT**: Hoeffding decomposition of $\Pi_{1,1}$; martingale-difference array; Lyapunov ($\gamma = 3$) and stability conditions in both regimes